

CS-302: Artificial Intelligence

LECTURE NO 6:

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Organization of Lecture

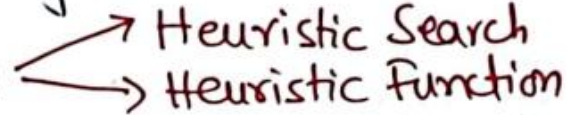
1. Informed Search Strategies

1. Introduction
2. Heuristic Search
3. Heuristic Function
4. Best First Search

1. Introduction (1/3)

↳ Information about GOAL state is present. (i.e, In Uninformed Search, the information about GOAL state is not known).

↳ Better than Uninformed Search.

↳ Informed Search 

↳ finds optimal solution to reach GOAL state.

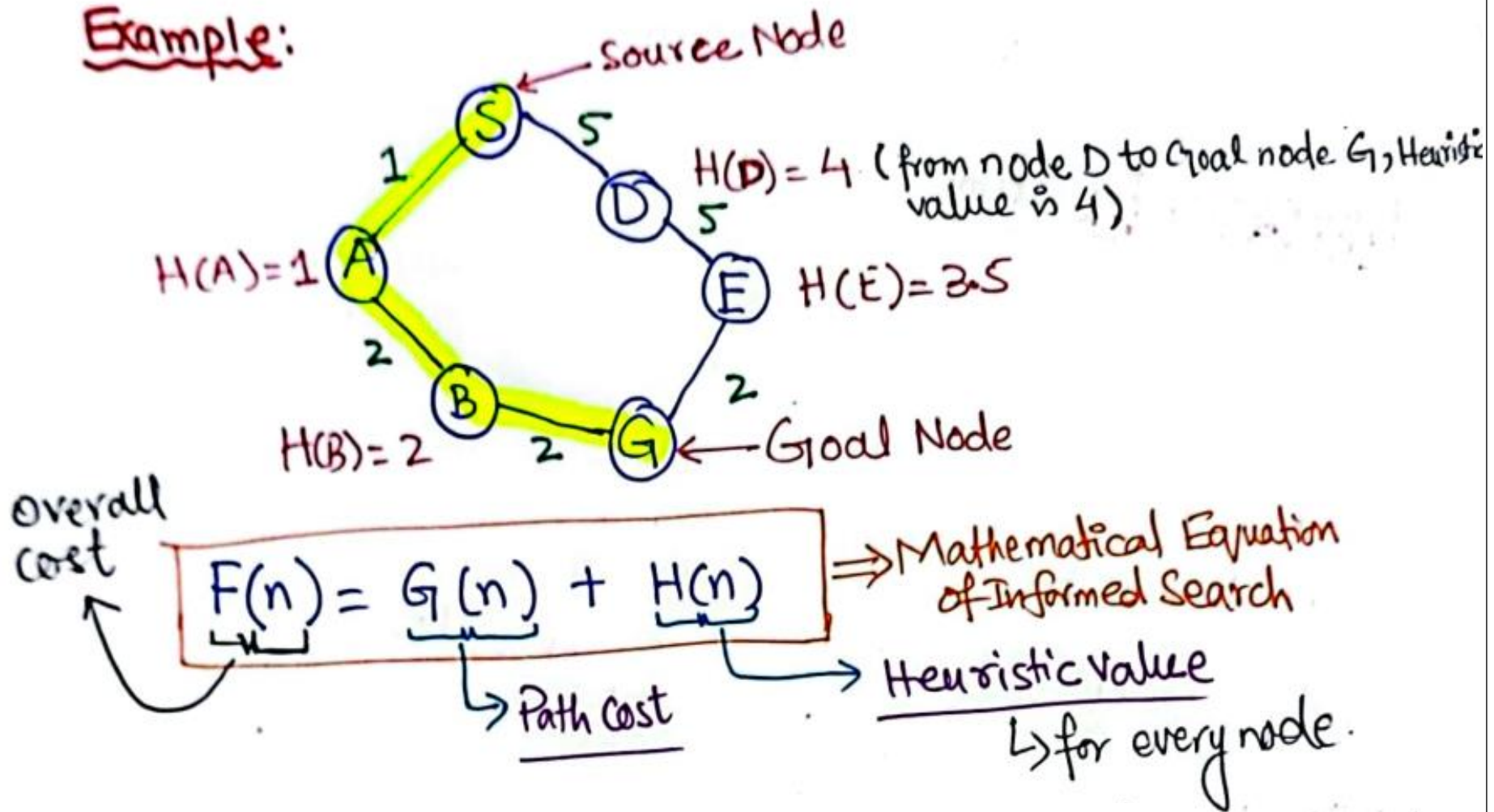


↳ It is a search which tries to reduce amount of search that must be done by making intelligent choices for the nodes that are selected for expansion.

↳ Eg: 

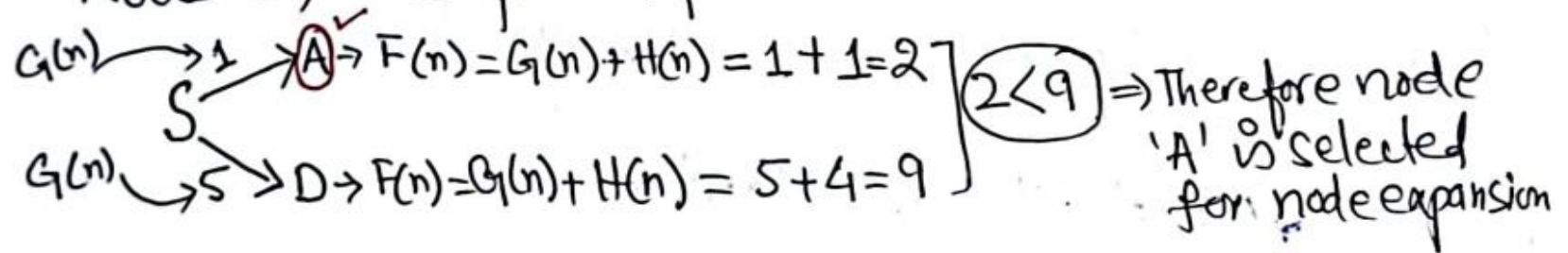
1. Introduction (2/3)

Example:



1. Introduction (3/3)

→ let : us say we want to go to Goal Node 'G' from Source Node 'S', The optimal path will be:



$$A \rightarrow B [F(n) = \overbrace{G(n)}^{1+2} + H(n) = \overbrace{(1+2)}^{2} + 2 = 5]$$

$B \rightarrow G$; Reached to Goal Node.

Total Path Cost (TPC)

$$TPC = S \xrightarrow{1} A \xrightarrow{2} B \xrightarrow{2} G$$

$$TPC = 1 + 2 + 2 = 5$$

$$\boxed{TPC = 5}$$

2. Heuristic Search

2) Heuristic Search:

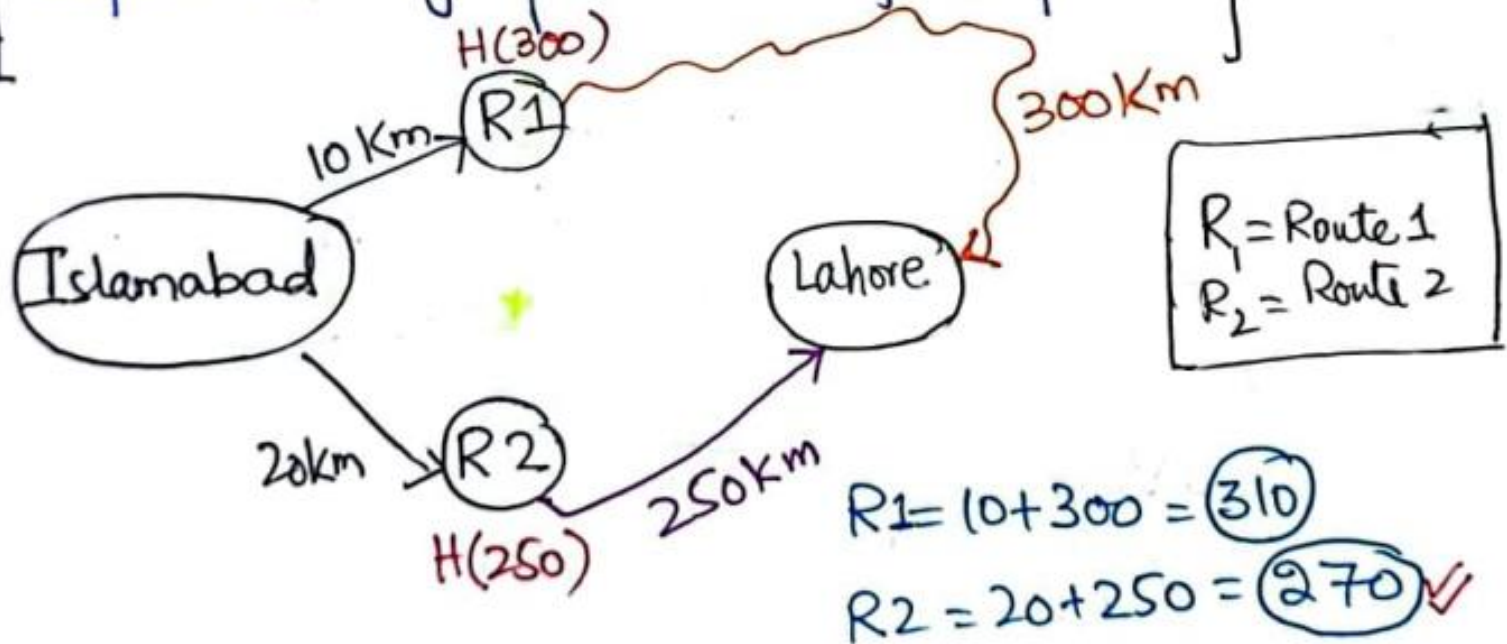
- ↳ Tries to optimize a problem using heuristic function.
- ↳ Optimization means trying to solve a problem in minimum cost / minimum number of steps.
- ↳ Heuristic Search & Heuristic Function are used in Informed Search.

3. Heuristic Function (1/5)

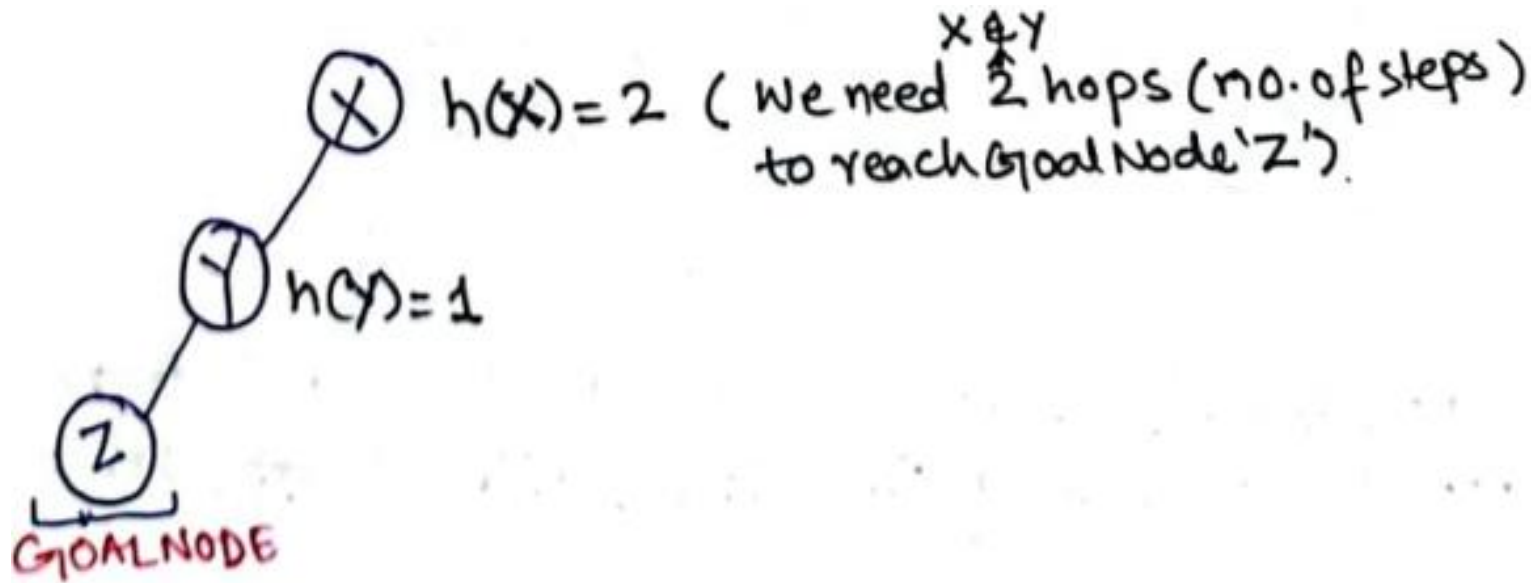
3) Heuristic Function:

It is a function ' $H(n)$ ' that gives an estimation on the cost of getting from node ' n ' to the GOAL state. } Estimated Value

Helps in selecting optimal node for expansion



3. Heuristic Function (2/5)



3. Heuristic Function (3/5)

Types of Heuristic Search

↳ (i) Admissible: In this Heuristic function, never overestimate the cost of reaching the Goal node. (Underestimate)

* $h(n)$ is always less than or equal to actual cost of lowest-cost path from node 'n' to goal

$$H(n) \leq H'(n)$$

$H(n) = \text{Estimated Cost}$ $H'(n) = \text{Actual Cost}$
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(ii) Non-Admissible: Overestimate $\Rightarrow H(n) > H'(n)$

3. Heuristic Function (4/5)

Example

$$\rightarrow H(B)=3, H(C)=4, H(D)=5$$

$$\rightarrow F(n) = H(n) + G(n)$$

$$\rightarrow \underline{B} \Rightarrow F(n) = 1+3=4 \checkmark$$

$$\rightarrow \underline{C} \Rightarrow F(n) = 1+4=5$$

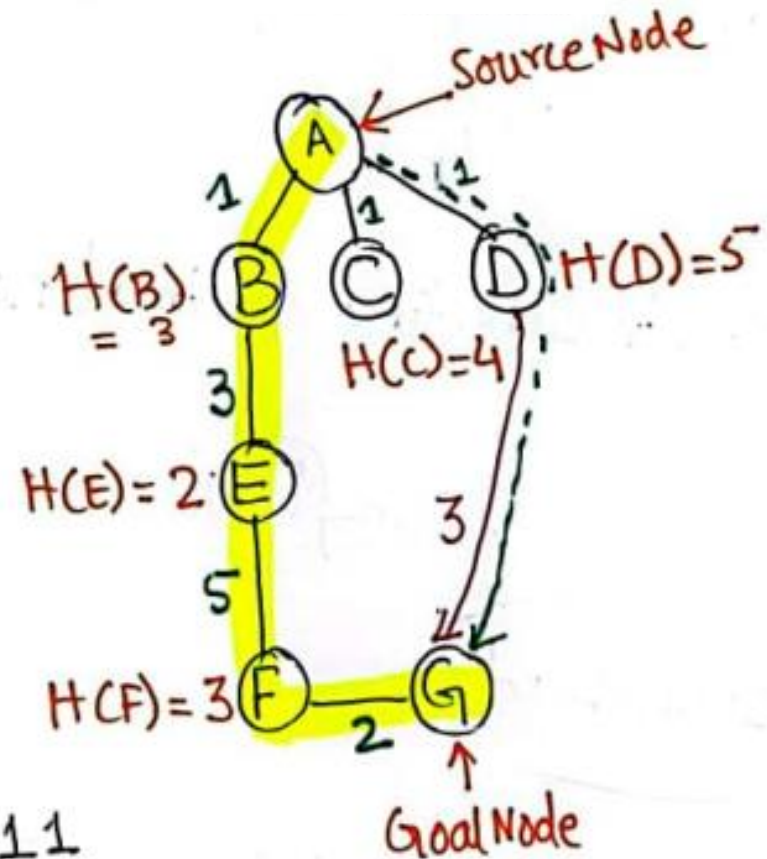
$$\rightarrow \underline{D} \Rightarrow F(n) = 1+5=6$$

B:-

$$\rightarrow \text{Actual Cost} = H'(B) = 1+3+5+2 = 11$$

$$4 \quad H(B)=3$$

$$\rightarrow H(B) < H(B') \Rightarrow 3 < 11 \Rightarrow \text{Admissible}$$



3. Heuristic Function (5/5)

→ In Case of E $\Rightarrow H(E) < H'(E) \Rightarrow 2 < 11$ (Admissible)

→ In case of F $\Rightarrow H(F) < H'(F) \Rightarrow 3 < 11$ (Admissible)

→ In Admissible Search, at each node $H(n) \leq H'(n)$

→ In Non-Admissible Search, If we had selected path $A \xrightarrow{1} D \xrightarrow{3} G$, we would have come up with $H(D) > H'(D) \Rightarrow 5 > 1+3 \Rightarrow 5 > 4$

4. Best First Search (1/6)

4) Best First Search:

- ↳ Uses evaluation Algorithm (function) to decide which adjacent node is most promising and then explore.
- ↳ Category of Heuristic or Informed Search.
- ↳ It is a GREEDY Search Algorithm.
- ↳ Priority Queue is used to store cost of nodes.
- ↳ It is a combination of BFS and DFS.

4. Best First Search (2/6)

Algorithm:

Priority Queue 'PQ' containing initial states → Sorted Order

Loop

if PQ = Empty Return FAIL

Else

NODE ← Remove-First(PQ)

if NODE = GOAL

Return Path from Initial to NODE

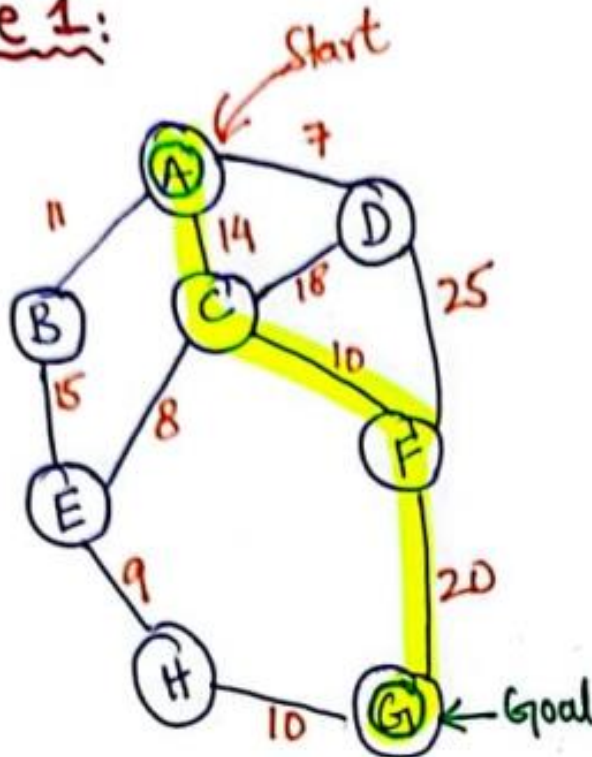
Else

Generate all Successors of NODE
and insert newly generate NODE
into PQ according to cost value.

END LOOP

4. Best First Search (3/6)

Example 1:



→ Usually given with Question
Straight line distance

$$A \rightarrow G_1 = 40$$

$$B \rightarrow G_1 = 32$$

$$C \rightarrow G_1 = 25$$

$$D \rightarrow G_1 = 35$$

$$E \rightarrow G_1 = 19$$

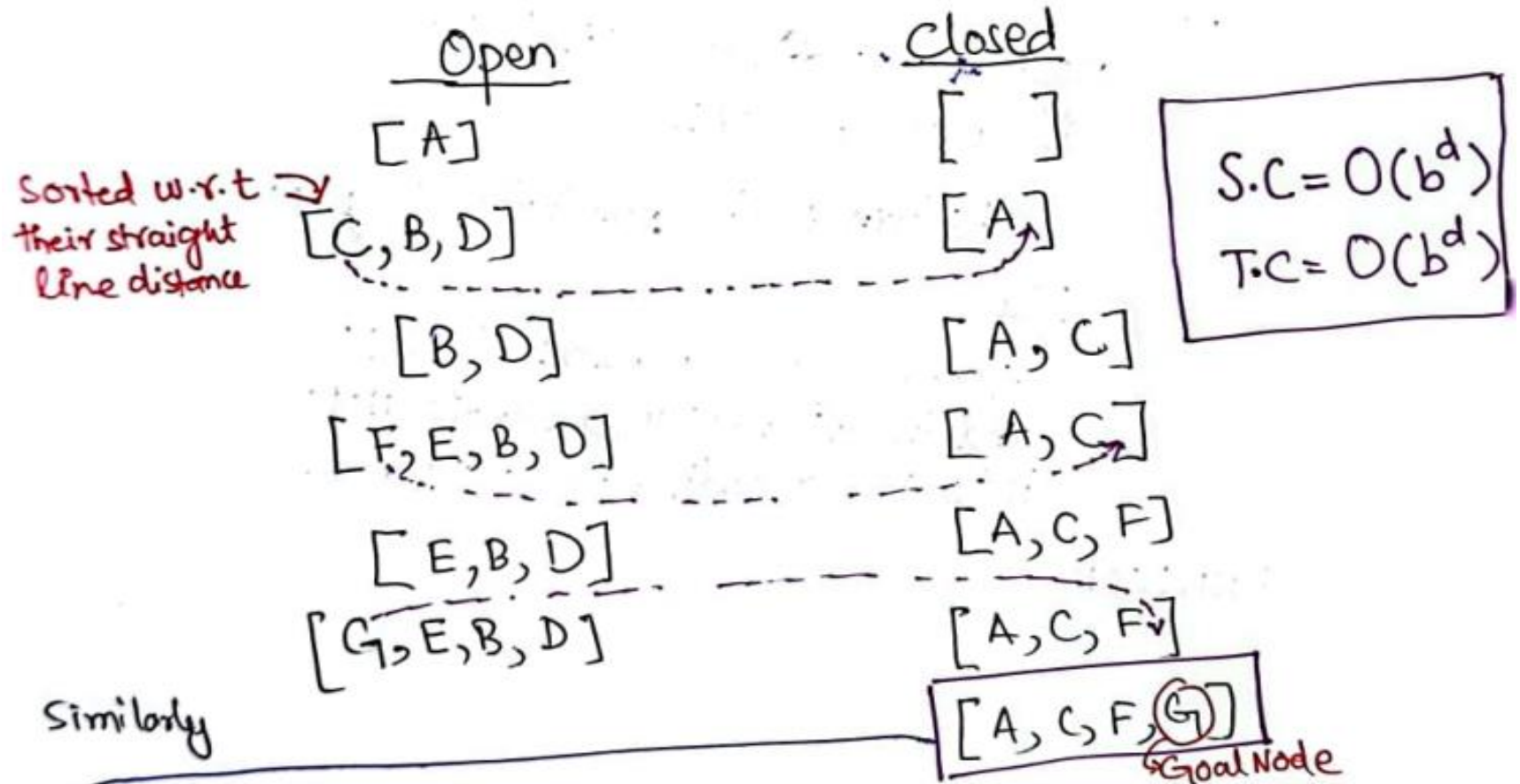
$$F \rightarrow G_1 = 17$$

$$G \rightarrow G_1 = 0$$

$$H \rightarrow G_1 = 10$$

→ We need to maintain two lists i.e., open List & Closed List.

4. Best First Search (4/6)



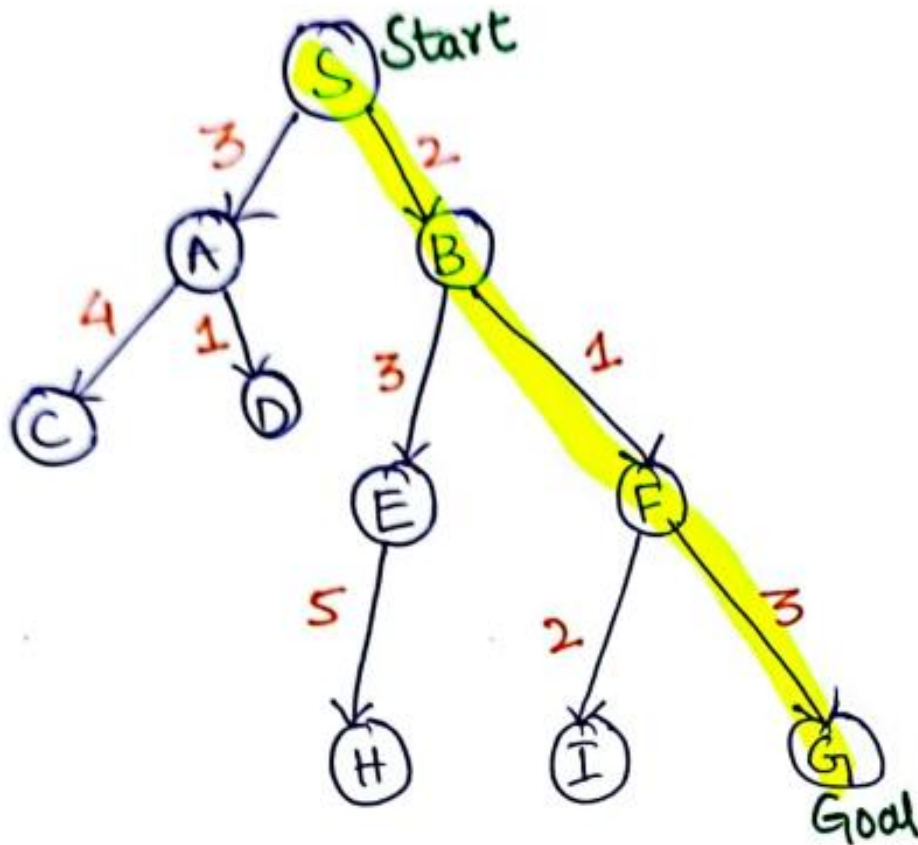
$$S.C = O(b^d)$$

$$T.C = O(b^d)$$

- Return this path i.e, $A \xrightarrow{14} C \xrightarrow{10} F \xrightarrow{20} G = 14 + 10 + 20 = 44$
- It might be possible that some other optimal path exists, so best first search will not always give you the best path

4. Best First Search (5/6)

Example 2:



Given:

<u>node(n)</u>	<u>H(n)</u>
A	12
B	4
C	7
D	3
E	8
F	2
H	4
I	9
S	13
G	0

4. Best First Search (6/6)

Open List

[S]

[B, A]

[F, E, A]

[G, I, E, A]

Closed List

[]

[S]

[S, B]

[S, B, F]

[S, B, F, G]

Goal Node

Return this path i.e., $S \xrightarrow{2} B \xrightarrow{1} F \xrightarrow{3} G = 2+1+3 = \boxed{6}$

References: Textbooks

1. *“Artificial Intelligence”, by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
2. *“Artificial Intelligence: A Modern Approach” by Stuart Russell and Peter Norvig, (2010), Prentice Hall, Chapter 1-27. page 1-1151.*
3. *“Computational Intelligence: A logical Approach”, by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *“Artificial Intelligence: Structures and Strategies for Complex Problem Solving”, by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *“AI: A new Synthesis”, by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *“Artificial Intelligence: Theory and Practice” by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*